

Unit 2: Python Control Flow, Functions & Modules • Topic 2

Loop Statements — For and While Loop

Repeating blocks of code using for and while loops, nested loops, and the loop-else clause.

Loops implement the **Iteration** building block of logic (Unit 1) — they let a block of code run repeatedly, either a fixed number of times or until a condition changes, instead of writing the same statement over and over. Python provides two loop statements: for and while.

2.1 The for Loop

A for loop iterates over a sequence (such as a range, a string, or a list) and executes its block once for **each item** in that sequence. It is the natural choice when you know, in advance, what you are iterating over (or how many times to repeat).

```
for variable in sequence:
    statement(s)
```

The range() Function

range() generates a sequence of numbers and is the most common way to control how many times a for-loop repeats.

Form	Produces	Example
range(stop)	0, 1, 2, ... stop-1	range(5) → 0,1,2,3,4
range(start, stop)	start, start+1, ... stop-1	range(2, 6) → 2,3,4,5
range(start, stop, step)	start, start+step, ... (before stop)	range(10, 0, -2) → 10,8,6,4,2

```
for i in range(1, 6):
    print(i, end=" ")
# Output: 1 2 3 4 5

for ch in "Python":
    print(ch, end="-")
# Output: P-y-t-h-o-n-
```

a for loop can iterate over a string too

A for loop can also iterate directly over other sequences such as lists, tuples, and dictionaries (covered in Units 3 & 4), e.g. for item in [10, 20, 30]: print(item).

2.2 The while Loop

A while loop repeats its block **as long as** a given condition remains True. It is the natural choice when you do *not* know in advance how many times the loop must run — the repetition depends on something that happens during the loop itself.

```
while condition:
    statement(s)    # must eventually make condition False, or it loops forever
```

```
count = 1
while count <= 5:
    print("Count is", count)
    count = count + 1    # IMPORTANT: update the variable, or the loop never ends
print("Loop finished")
```

COMMON PITFALL: Forgetting to update the loop-control variable (like count above) creates an **infinite loop** — the program will hang. Always make sure something inside a while loop moves the condition towards False.

2.3 for vs while — When to Use Which

Aspect	for loop	while loop
Best used when	Number of iterations is known / iterating a sequence	Repetition depends on a changing condition
Loop variable update	Handled automatically by the sequence/range	Must be updated manually inside the loop
Typical example	Printing a multiplication table (1 to 10)	Reading input until user types 'exit'

2.4 Nested Loops

A loop placed inside another loop is called a **nested loop**. For each single iteration of the outer loop, the entire inner loop runs to completion. Nested loops are common for working with grids, patterns, and tables.

```
for i in range(1, 4):          # outer loop: 3 iterations
    for j in range(1, 3):      # inner loop: 2 iterations each time
        print(f"i={i}, j={j}")
# Runs the inner loop's print statement 3 x 2 = 6 times in total
```

Worked Example: Printing a Number Pattern

```
for i in range(1, 5):
    for j in range(1, i + 1):
        print(j, end=" ")
    print()    # move to a new line after each row
# Output:
# 1
# 1 2
# 1 2 3
# 1 2 3 4
```

2.5 The else Clause with Loops

Python allows an optional else block after a for or while loop. It executes **only if the loop completes normally** (i.e. it is NOT skipped because of a break — see Topic 3). This is a Python-specific feature, useful for search-and-report patterns.

```
numbers = [3, 7, 9, 15]
for n in numbers:
    if n % 2 == 0:
        print("Found an even number:", n)
        break
    else:
        print("No even number found")    # runs because the loop never hit 'break'
```

Practice Questions

A. Multiple Choice Questions

- What does `range(2, 10, 2)` produce?
(A) 2,4,6,8,10 (B) 2,4,6,8 (C) 2,3,4,...,9 (D) 0,2,4,6,8
- Which loop is best when the number of iterations is not known in advance?
(A) for loop (B) while loop (C) Both are equally suited (D) Neither
- What is the main risk of a while loop if the condition variable is never updated?
(A) `SyntaxError` (B) It runs exactly once (C) Infinite loop (D) It skips the loop entirely
- In a nested loop with an outer loop of 3 iterations and an inner loop of 4 iterations, how many times does the innermost statement execute in total?
(A) 3 (B) 4 (C) 7 (D) 12
- The `else` clause after a for loop executes when:
(A) The loop is skipped entirely (B) The loop completes without hitting a `break` (C) The loop hits a `break` (D) Never, it is invalid syntax

B. True / False

- `range(5)` produces the numbers 1 to 5.
- A for loop in Python can iterate directly over the characters of a string.
- A while loop's condition is checked before every iteration, including the first one.
- Nested loops mean a loop placed inside another loop's body.

C. Short Answer Questions

- Explain the three forms of the `range()` function with one example each.
- What is a nested loop? Give a real-world scenario where a nested loop is a natural fit.
- Differentiate between a for loop and a while loop, and state when each should be preferred.
- What causes an infinite while loop? How can it be avoided?

D. Long Answer / Descriptive Questions

- Write a Python program using nested for loops to print the following pattern for `n=5`:

```
*
**
***
****
*****
```
- Explain, with a program, how a while loop can be used to repeatedly ask the user for a number until they enter 0 (which stops the loop), printing the running total of all numbers entered so far.

E. Programming Problems

- Write a Python program to print the multiplication table (1 to 10) of a number entered by the user, using a for loop.
- Write a Python program using a while loop to find the sum of digits of a given number (e.g. 1234 → 1+2+3+4 = 10).
- Write a Python program using nested loops to print all pairs (i, j) where $1 \leq i \leq 3$ and $1 \leq j \leq 3$ and i is not equal to j.
- Write a Python program that uses a for-else construct to check whether a given number is prime (hint: loop from 2 to n-1 and use `break` when a factor is found).

Answer Key (A & B): MCQ: 1-B, 2-B, 3-C, 4-D, 5-B | T/F: 6-False, 7-True, 8-True, 9-True